1. **Write a query that lists each order number followed by the name of the customer who made the order.**

```
d6_96728_shaurya> SELECT o.onum, c.cname FROM orders o
    -> RIGHT JOIN customers c ON c.cnum = o.cnum;
+------+----------+
| onum | cname    |
+------+----------+
| 3003 | Hoffman  |
| 3009 | Giovanni |
| 3005 | Liu      |
| 3010 | Grass    |
| 3007 | Grass    |
| 3011 | Clemens  |
| 3008 | Clemens  |
| 3006 | Cisneros |
| 3001 | Cisneros |
| 3002 | Pereira  |
+------+----------+
10 rows in set (0.007 sec)
```

2. **Write a query that gives the names of both the salesperson and the customer for each order along with the order number.**

```
d6_96728_shaurya> SELECT o.onum, c.cname, s.sname from customers c
    -> LEFT JOIN salespeople s ON s.snum = c.snum
    -> Join orders o ON o.cnum = c.cnum;
+------+----------+---------+
| onum | cname    | sname   |
+------+----------+---------+
| 3001 | Cisneros | Rifkin  |
| 3003 | Hoffman  | Peel    |
| 3002 | Pereira  | Motika  |
| 3005 | Liu      | Serres  |
| 3006 | Cisneros | Rifkin  |
| 3009 | Giovanni | Axelrod |
| 3007 | Grass    | Serres  |
| 3008 | Clemens  | Peel    |
| 3010 | Grass    | Serres  |
| 3011 | Clemens  | Peel    |
+------+----------+---------+
10 rows in set (0.007 sec)
```

3. **Write a query that produces all customers serviced by salespeople with a commission above 12%. Display the customer's name, the salesperson's name, and the salesperson's rate of commission.**

```
d6_96728_shaurya> SELECT c.cname, s.sname, s.comm FROM customers c
    -> LEFT JOIN salespeople s ON s.snum = c.snum
    -> WHERE s.comm>0.12;
+----------+--------+------+
| cname    | sname  | comm |
+----------+--------+------+
| Liu      | Serres | 0.13 |
| Grass    | Serres | 0.13 |
| Cisneros | Rifkin | 0.15 |
+----------+--------+------+
3 rows in set (0.006 sec)
```

4. **Write a query that calculates the commission earned by each salesperson for orders placed by customers with a rating above 100.**

```
d6_96728_shaurya> select o.onum, s.comm, s.comm * o.amt from orders o
    -> left join salespeople s ON s.snum = o.snum
    -> right join customers c ON c.cnum = o.cnum
    -> WHERE c.rating > 100;
+------+------+----------------+
| onum | comm | s.comm * o.amt |
+------+------+----------------+
| 3009 | 0.10 |       171.3230 |
| 3005 | 0.13 |       670.8585 |
| 3010 | 0.13 |        40.2935 |
| 3007 | 0.13 |         9.8475 |
| 3006 | 0.15 |       164.7240 |
| 3001 | 0.15 |         2.8035 |
+------+------+----------------+
6 rows in set (0.007 sec)
```

5. **Write a query that produces all pairs of salespeople who are living in the same city. Exclude combinations of salespeople with themselves as well as duplicate rows with the order reversed.**

This is correct:

select s.sname, o.sname, s.city, o.city from salespeople s
JOIN salespeople o ON s.snum = o.snum where s.city = o.city;

Select a.sname, b.sname from salespeople a, salespeople b where a.sname > b.sname and a.city = b.city;

This is wrong:

Select s.sname, s.city, o.sname, o.city from salespeople s
inner join salespeople o ON s.snum = o.snum
where s.city = o.city;

sname	city	sname	city
Peel	London	Peel	London
Serres	San Jose	Serres	San Jose
Motika	London	Motika	London
Rifkin	Barcelona	Rifkin	Barcelona
Axelrod	New York	Axelrod	New York

5 rows in set (0.007 sec)

In both of your attempts, I joined the table on the same unique ID (snum or sname). This essentially "match every person with themselves," which is why no pairs of *different* people.
To get all pairs in the same city, you need to join based on the city column instead of the ID.
s.sname > o.sname: It prevents duplicates like (Peel, Motika) and (Motika, Peel) because only one of those names will be alphabetically "smaller" than the other

```
d6_96728_shaurya> Select s.sname, s.city, o.sname, o.city from salespeople s

    -> inner join salespeople o ON s.city = o.city
    -> where s.city = o.city;
+---------+-----------+---------+-----------+
| sname   | city      | sname   | city      |
+---------+-----------+---------+-----------+
| Motika  | London    | Peel    | London    |
| Peel    | London    | Peel    | London    |
| Serres  | San Jose  | Serres  | San Jose  |
| Motika  | London    | Motika  | London    |
| Peel    | London    | Motika  | London    |
| Rifkin  | Barcelona | Rifkin  | Barcelona |
| Axelrod | New York  | Axelrod | New York  |
+---------+-----------+---------+-----------+
7 rows in set (0.007 sec)

d6_96728_shaurya> Select s.sname, s.city, o.sname, o.city from salespeople s

    -> inner join salespeople o ON s.city = o.city
    -> where s.city > o.city;
Empty set (0.006 sec)

d6_96728_shaurya> Select s.sname, s.city, o.sname, o.city from salespeople s

    -> inner join salespeople o ON s.city = o.city
    -> where s.sname != o.sname AND s.city = o.city;
+--------+--------+--------+--------+
| sname  | city   | sname  | city   |
+--------+--------+--------+--------+
| Motika | London | Peel   | London |
| Peel   | London | Motika | London |
+--------+--------+--------+--------+
2 rows in set (0.007 sec)
```

```
d6_96728_shaurya> Select s.sname, s.city, o.sname, o.city from salespeople s

    -> inner join salespeople o ON s.city = o.city
    -> where s.sname > o.sname AND s.city = o.city;
+-------+--------+--------+--------+
| sname | city   | sname  | city   |
+-------+--------+--------+--------+
| Peel  | London | Motika | London |
+-------+--------+--------+--------+
1 row in set (0.007 sec)

d6_96728_shaurya>
```